

SOFTWARE REQUIREMENTS SPECIFICATION

Measurement Process Simulator

Prepared by:

Elifnaz AKKAYNAK

1. Introduction

1.1 Purpose

This document describes the software requirements of the ISO 15939 Measurement Process Simulator. The purpose of this system is to simulate the software measurement process defined in the ISO/IEC 15939 standard.

1.2 Scope

The system is a Java Swing desktop application that simulates the ISO/IEC 15939 measurement process including profile creation, defining quality dimensions, planning metrics, collecting data, and analyzing results.

1.3 Definitions

Term	Definition
Metric	A measurable value used to evaluate a system
Dimension	A group of related metrics
Score	A calculated value between 1 and 5
Scenario	A predefined dataset for evaluation

1.4 References

- [1] ISO/IEC 15939: Systems and software engineering — Measurement process
- [2] SENG 272 Course Assignment Description

1.5 Overview

This document is organized as follows:

- **Section 2** provides an overall description
- **Section 3** defines system features and requirements

- **Section 4** describes non-functional requirements

2. Overall Description

2.1 Product Perspective

The system is a standalone desktop application developed using Java Swing. It does not require any external libraries or database connection.

2.2 User Characteristics

- Users have basic computer knowledge
- No programming experience is required

2.3 Assumptions

ID	Requirement
ASM-01	All data is hard-coded
ASM-02	The system runs locally
ASM-03	No internet connection is required

2.4 Constraints

ID	Requirement
CON-01	No external libraries are allowed
CON-02	No database usage
CON-03	The application must run from command line
CON-04	All data must be hard-coded

3. System Features

3.1 Profile Step

The system shall allow the user to enter:

- Username
- School
- Session Name

Requirements:

ID	Requirement
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FR-PRF-01	All fields must be filled
FR-PRF-02	If any field is empty, a warning message must be displayed
FR-PRF-03	The system must prevent moving to the next step

3.2 Define Quality Step

The system shall allow the user to select:

- Quality Type (Product / Process)
- Mode (Health / Education / Custom)
- Scenario

Requirements:

ID	Requirement
FR-DQS-01	Only one option can be selected at a time
FR-DQS-02	Scenario list changes based on selected mode
FR-DQS-03	User must select all options before proceeding

3.3 Plan Measurement Step

The system shall display predefined data in a table format.

Displayed information:

- Dimension name
- Metric name
- Coefficient
- Direction
- Range
- Unit

Requirements:

ID	Requirement
FR-PMS-01	Table must be read-only
FR-PMS-02	Data must be loaded based on selected scenario

3.4 Collect Data Step

The system shall display metric values and calculate scores.

Score Calculation:

$$score = 1 + \frac{value - min}{max - min} \times 4$$

$$score = 5 - \frac{value - min}{max - min} \times 4$$

Requirements:

ID	Requirement
FR-CDS-01	Scores must be between 1.0 and 5.0
FR-CDS-02	Values must be rounded to nearest 0.5
FR-CDS-03	Data is predefined (hard-coded)

3.5 Analyse Step

The system shall analyze collected data.

Dimension Score Calculation:

$$dimensionScore = \frac{\sum(metricScore \times metricCoefficient)}{\sum(metricCoefficient)}$$

Features:

- Display scores using progress bars
- Identify lowest scoring dimension
- Calculate gap value

Gap Analysis Output:

- Dimension name
- Score
- Gap value (5 - score)
- Quality level:
 - Excellent
 - Good
 - Needs Improvement
 - Poor

4. Non-Functional Requirements

ID	Requirement
NFR-01	The system must be developed using Java SE 17
NFR-02	The GUI must be implemented using Java Swing
NFR-03	The system must follow MVC architecture
NFR-04	The system must be responsive and user-friendly
NFR-05	Code must be readable and well-structured

